Spruce Budworm Outbreaks and Hysteresis

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Notation: Budworm variables and parameters

Let $N(t) \geq 0$ be the budworm population, and let

$$\begin{array}{r|c} R > 0 & \text{intrinsic growth rate} \\ K > 0 & \text{carrying capacity} \\ A > 0 & \text{half-saturation population} \\ B > 0 & \text{maximum predation rate} \end{array}$$

Here K is the carrying capacity of the logistic growth term; equivalently, it is the carrying capacity the population would have in the absence of predation.

Idea: Growth minus predation

Model the population rate as logistic growth minus saturating predation:

$$\boxed{\dot{N} = RN\left(1 - \frac{N}{K}\right) - \frac{BN^2}{A^2 + N^2}}.$$

Observation: Saturating predation response

The predation term is quadratic near extinction, reaches half of its maximum at $N = A$, and saturates at B:

$$\frac{BN^2}{A^2 + N^2} = O(N^2) \quad (N \to 0), \qquad \frac{BA^2}{A^2 + A^2} = \frac{B}{2},$$

$$\frac{BN^2}{A^2 + N^2} \longrightarrow B \qquad (N \to \infty).$$

Question: How is the budworm model nondimensionalized?

Define

$$x = \frac{N}{A}, \qquad \tau = \frac{B}{A}t, \qquad r = \frac{RA}{B}, \qquad k = \frac{K}{A}.$$

Derive the dimensionless equation.

Solution. Since $N = Ax$ and $\tau = (B/A)t$,

$$\dot{N} = A\frac{dx}{dt} = A\frac{dx}{d\tau}\frac{d\tau}{dt} = B\frac{dx}{d\tau}.$$

Also,

$$RN\left(1 - \frac{N}{K}\right) = RAx\left(1 - \frac{x}{k}\right)$$

and

$$\frac{BN^2}{A^2 + N^2} = B\frac{x^2}{1 + x^2}.$$

Division by B gives

$$\boxed{\frac{dx}{d\tau} = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}}.$$

Observation: Extinction is unstable

For

$$F(x) = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2},$$

$F(0) = 0$ and $F'(0) = r > 0$. Hence extinction is unstable from the physical side $x \geq 0$.

Visualization: Positive equilibria as line–curve intersections

On the physical half-line $x \geq 0$, define

$$L(x) = r\left(1 - \frac{x}{k}\right), \qquad H(x) = \frac{x}{1 + x^2}.$$

For $x > 0$, the equilibrium equation is $L(x) = H(x)$.

	$L(x)$	$H(x)$
vertical intercept	r	0
horizontal intercept	k	0
derivative	$-r/k$	$\dfrac{1 - x^2}{(1 + x^2)^2}$
maximum	r at $x = 0$	$1/2$ at $x = 1$
$x \to \infty$	$-\infty$	0

The intersections give every positive equilibrium; the equilibrium $x = 0$ must be added separately.

Proposition: Positive multiple-equilibrium locus

Let $r, k > 0$ and

$$F(x; r, k) = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}.$$

A positive equilibrium $x = s$ is multiple if and only if $s > 1$ and

$$\boxed{r = \frac{2s^3}{(1 + s^2)^2}, \qquad k = \frac{2s^3}{s^2 - 1}}.$$

Proof. For $s > 0$, write

$$F(s; r, k) = s\left[r\left(1 - \frac{s}{k}\right) - \frac{s}{1 + s^2}\right].$$

The equations $F(s; r, k) = F_x(s; r, k) = 0$ are equivalent to

$$r\left(1 - \frac{s}{k}\right) = \frac{s}{1 + s^2}, \qquad \frac{r}{k} = \frac{s^2 - 1}{(1 + s^2)^2}.$$

Solving gives the displayed formulas. The condition $k > 0$ is equivalent to $s > 1$. □

Definition: Ordinary semicubical cusp of a plane curve

Let $\Gamma : I \to \mathbb{R}^2$ be three times continuously differentiable. The curve Γ has an **ordinary semicubical cusp** at $s_c \in I$ if

$$\Gamma'(s_c) = \mathbf{0}, \qquad \det\Big(\Gamma''(s_c), \Gamma'''(s_c)\Big) \neq 0.$$

Proposition: Cusp of the fold locus

For $s > 1$, define

$$k(s) = \frac{2s^3}{s^2 - 1}, \qquad r(s) = \frac{2s^3}{(1 + s^2)^2}.$$

The two regular arms of this curve meet in an ordinary semicubical cusp at

$$s_c = \sqrt{3}, \qquad k_c = 3\sqrt{3}, \qquad r_c = \frac{3\sqrt{3}}{8}.$$

Proof. Differentiation gives

$$k'(s) = \frac{2s^2(s^2 - 3)}{(s^2 - 1)^2}, \qquad r'(s) = \frac{2s^2(3 - s^2)}{(1 + s^2)^3}.$$

Both derivatives vanish only at $s = \sqrt{3}$. At that point,

$$(k'', r'') = \left(3\sqrt{3}, -\frac{3\sqrt{3}}{16}\right), \qquad (k''', r''') = \left(-21, \frac{3}{4}\right),$$

and the determinant of these vectors is $-27\sqrt{3}/16 \neq 0$. □

Proposition: Ordinary saddle-nodes on the budworm fold locus

Let
$$F(x;r,k) = rx\left(1-\frac{x}{k}\right) - \frac{x^2}{1+x^2}, \qquad r,k>0.$$
Suppose that $s>1$, $s\neq\sqrt{3}$, and
$$r = \frac{2s^3}{(1+s^2)^2}, \qquad k = \frac{2s^3}{s^2-1}.$$
Then $x=s$ is an ordinary saddle-node.

Proof. Set
$$q(x) = r\left(1-\frac{x}{k}\right) - \frac{x}{1+x^2}, \qquad F(x;r,k) = xq(x).$$
On the multiple-equilibrium locus, $q(s)=q'(s)=0$, and
$$q''(s) = -\frac{2s(s^2-3)}{(1+s^2)^3}.$$
Thus $F_{xx}(s;r,k) = sq''(s)\neq 0$. Moreover,
$$F_r(s;r,k) = s\left(1-\frac{s}{k}\right) = \frac{1+s^2}{2s} > 0.$$
These are the nondegeneracy and parameter-transversality conditions for an ordinary saddle-node. □

Proposition: Generic attracting cusp equilibrium in the budworm model

Let
$$F(x;r,k) = rx\left(1-\frac{x}{k}\right) - \frac{x^2}{1+x^2}.$$
At
$$x_c = \sqrt{3}, \qquad k_c = 3\sqrt{3}, \qquad r_c = \frac{3\sqrt{3}}{8},$$
x_c is a generic triple equilibrium that attracts from both sides.

Proof. Define
$$q(x) = r_c\left(1-\frac{x}{k_c}\right) - \frac{x}{1+x^2}, \qquad F(x;r_c,k_c) = xq(x).$$
At $x=x_c$,
$$F(x_c;r_c,k_c) = F_x(x_c;r_c,k_c) = F_{xx}(x_c;r_c,k_c) = 0,$$
while
$$F_{xxx}(x_c;r_c,k_c) = -\frac{3\sqrt{3}}{16}\neq 0, \qquad \det\begin{pmatrix} F_k & F_r \\ F_{xk} & F_{xr}\end{pmatrix}\Bigg|_{(x_c;r_c,k_c)} = -\frac{\sqrt{3}}{24}\neq 0.$$
Hence the triple root satisfies the generic cusp conditions. Finally,
$$q(x) = -\frac{1}{32}(x-x_c)^3 + o(|x-x_c|^3) \qquad (x\to x_c),$$
so $F(x;r_c,k_c)$ points toward x_c from both sides. □

Visualization: Budworm stability diagram

Let

$$\mathcal{F} = \{(k(x), r(x)) : x > 1\}$$

be the fold locus. Its two arms correspond to

$$1 < x < \sqrt{3} \qquad \text{and} \qquad x > \sqrt{3}.$$

The two arms bound a region in the positive (k, r)-parameter plane. In the table below, **inside the cusp** means strictly between these arms, and **outside the cusp** means in their complement away from the fold locus.

parameter location	positive equilibria	equilibrium types
outside the cusp	1	one asymptotically stable
on $\mathcal{F} \setminus \{(k_c, r_c)\}$	2 distinct, one double	one asymptotically stable simple; one half-stable double
(k_c, r_c)	1 triple	one nonhyperbolic attracting
inside the cusp	3	two asymptotically stable; one unstable

For $k > k_c$, the parameter path $k =$ constant intersects both fold arms, so varying r passes through refuge-only, bistable, and outbreak-only regions.

Definition: Refuge level

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If a is asymptotically stable, then a is the **refuge level**.

Definition: Outbreak threshold

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If b is unstable and separates the basins of a and c, then b is the **outbreak threshold**.

Definition: Outbreak level

Suppose a scalar population model has three positive equilibria

$$0 < a < b < c.$$

If c is asymptotically stable, then c is the **outbreak level**.

Recall: Bistability

On a specified state space, **bistability** means that the system has exactly two distinct asymptotically stable equilibria in that state space.

Recall: Hysteresis

Hysteresis is path dependence in which the observed state–parameter response does not retrace the same path when the direction of parameter variation is reversed.

Observation: Fold-induced hysteresis in the budworm model

Suppose a quasistatic parameter sweep begins on an asymptotically stable budworm equilibrium and crosses the bistable region. Increasing and decreasing sweeps can then track different branches and jump at different fold values.

Proposition: Stability and basins in the three-root regime

Assume $r, k > 0$ and that the positive equilibria are $0 < a < b < c$. Then a and c are asymptotically stable, b is unstable, and

$$\begin{array}{r|c} 0 < x_0 < b & x(\tau) \to a \\ x_0 = b & x(\tau) = b \\ x_0 > b & x(\tau) \to c. \end{array}$$

The initial condition $x_0 = 0$ remains at the unstable extinction equilibrium.

Proof. Write the vector field as

$$F(x) = x\left[r\left(1 - \frac{x}{k}\right) - \frac{x}{1+x^2}\right].$$

For small $x > 0$, $F(x) > 0$. For sufficiently large x, $F(x) < 0$. Since the three positive roots are simple, the sign alternates:

x	$(0, a)$	(a, b)	(b, c)	(c, ∞)
$F(x)$	$+$	$-$	$+$	$-$.

The vector field is continuously differentiable, $x = 0$ is an equilibrium solution, and $F(x) < 0$ for all sufficiently large x. Hence every positive solution is unique, remains positive, and is bounded forward in time. The sign table makes it monotone whenever it lies between consecutive equilibria. Therefore it converges to the unique equilibrium indicated by the arrows in its interval. $\square$

Question: What is the second fold for a specified carrying capacity?

Verify the fold point at $x = 2$ and find the other fold with the same k.

Solution. At $x = 2$,

$$k = \frac{2(2)^3}{2^2 - 1} = \frac{16}{3}, \qquad r = \frac{2(2)^3}{(1 + 2^2)^2} = \frac{16}{25}.$$

The intersection and slope equations are

$$\frac{16}{25}\left(1 - \frac{2}{16/3}\right) = \frac{2}{5} = \frac{2}{1 + 2^2}$$

and

$$-\frac{16/25}{16/3} = -\frac{3}{25} = \frac{1 - 2^2}{(1 + 2^2)^2}.$$

For the other tangency, solve

$$\frac{2x^3}{x^2 - 1} = \frac{16}{3}.$$

Then

$$3x^3 - 8x^2 + 8 = (x - 2)(3x^2 - 2x - 4) = 0,$$

so the other admissible root is

$$x = \frac{1 + \sqrt{13}}{3} \approx 1.53518.$$

Its parameter value is

$$r = \frac{2x^3}{(1 + x^2)^2} = \frac{304 + 416\sqrt{13}}{2809} \approx 0.64219.$$

Idea: Slow forest growth gives a parameter path

Let S be average tree size. Hold R and B fixed, and suppose

$$K = K'S, \qquad A = A'S,$$

where $K', A' > 0$ are constant. Then

$$k = \frac{K}{A} = \frac{K'}{A'}$$

is constant, while

$$r = \frac{RA}{B} = \frac{RA'}{B}S$$

increases with S. Forest maturation therefore follows the fixed-k path $k = K'/A'$ toward larger r. If the refuge branch reaches its fold, it disappears and the population moves to the outbreak branch.

Visualization: Empirical parameter scale reported with the model

forest state	k	r
young	≈ 300	$< 1/2$
mature	≈ 300	≈ 1

The reported recovery time of the fir forest after replacement by birch is approximately

$$50\text{–}100 \text{ years.}$$

These are empirical estimates used to locate a parameter path, not deductions from the dimensionless ODE.

Idea: Scope of the model

The outbreak mechanism is conditional on

$$\text{logistic growth} + \text{saturating predation} + \text{separation of insect and forest time scales.}$$

Spatial dispersal, heterogeneous foliage, and the slow tree dynamics are not state variables in the scalar model.